

## Lab 3: Relational Algebra

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### Question 1

*Find the first and last names of persons who have made at least one equipment reservation.*

#### Step 1: Join Person and Reservation

First, I join **Person** and **Reservation** using the person ID and the ID of the person who made the reservation. This allows us to associate each reservation with the person who made it.

$$R_1 = \text{Person} \bowtie_{\text{Person.pid}=\text{Reservation.madeByPid}} \text{Reservation}$$

#### Step 2: Project the first and last names

From the resulting relation, I only need the first and last names.

$$\pi_{\text{first name, last name}}(R_1)$$

Therefore, the final expression is:

$$\pi_{\text{first name, last name}}(\text{Person} \bowtie_{\text{Person.pid}=\text{Reservation.madeByPid}} \text{Reservation})$$

### Question 2

*Find the IDs of persons who are either attached to a laboratory or act as PI of at least one research project.*

### Step 1: Find persons attached to a laboratory

The relation `Attached` already contains the IDs of persons attached to laboratories, so I project the `pid` attribute.

$$R_1 = \pi_{\text{pid}}(\text{Attached})$$

### Step 2: Find persons who act as PI

First, I select the persons whose role is PI.

$$R_2 = \sigma_{\text{role=PI}}(\text{Participates})$$

Then, I project their IDs.

$$R_3 = \pi_{\text{pid}}(R_2)$$

### Step 3: Combine the two cases

Since the question asks for persons who satisfy either condition, I use the union operation.

$$\boxed{R_1 \cup R_3}$$

Therefore, the final expression is:

$$\boxed{\pi_{\text{pid}}(\text{Attached}) \cup \pi_{\text{pid}}(\sigma_{\text{role=PI}}(\text{Participates}))}$$

## Question 3

*Find the IDs of laboratories located in the building “College of Computing” or containing at least one equipment unit whose model belongs to the category “CPU Server”.*

### Step 1: Find laboratories in the College of Computing

First, I select the laboratories whose building is “College of Computing”.

$$R_1 = \sigma_{\text{building=“College of Computing”}}(\text{Laboratory})$$

Then, I project their laboratory IDs.

$$R_2 = \pi_{\text{labId}}(R_1)$$

## Step 2: Join equipment units with their models

To determine the category of each equipment unit, I join `EquipmentUnit` and `EquipmentModel` using `modelId`.

$$R_3 = \text{EquipmentModel} \bowtie_{\text{modelId}} \text{EquipmentUnit}$$

## Step 3: Keep only CPU Server models

I select the equipment models belonging to the “CPU Server” category.

$$R_4 = \sigma_{\text{category}=\text{“CPU Server”}}(R_3)$$

## Step 4: Find the laboratory IDs

I project the laboratory IDs.

$$R_5 = \pi_{\text{labId}}(R_4)$$

## Step 5: Combine the two cases

Since the question uses “or”, I take the union of the two sets of laboratory IDs.

$$\boxed{R_2 \cup R_5}$$

Therefore, the final expression is:

$$\boxed{\begin{aligned} &\pi_{\text{labId}}(\sigma_{\text{building}=\text{“College of Computing”}}(\text{Laboratory})) \\ &\cup \pi_{\text{labId}}(\sigma_{\text{category}=\text{“CPU Server”}}(\text{EquipmentModel} \bowtie_{\text{modelId}} \text{EquipmentUnit})) \end{aligned}}$$

## Question 4

*Find the IDs of laboratories that contain at least one equipment unit whose model belongs to the category “CPU Server” and at least one equipment unit whose model belongs to the category “GPU Server”.*

For this question, I need to find laboratories that are associated with both required categories. I use relational division for this.

## Step 1: Find the required categories

First, I select the equipment models that belong to either “CPU Server” or “GPU Server”.

$$R_1 = \sigma_{\text{category}=\text{“CPU Server”} \vee \text{category}=\text{“GPU Server”}}(\text{EquipmentModel})$$

Then, I project only the category attribute.

$$\text{categories} = \pi_{\text{category}}(R_1)$$

Thus, `categories` contains the two categories that every laboratory must have.

## Step 2: Join equipment units with their models

I join `EquipmentUnit` and `EquipmentModel` using their common `modelId`. This allows me to associate each equipment unit with its model.

$$R_2 = \text{EquipmentUnit} \bowtie_{\text{modelId}} \text{EquipmentModel}$$

## Step 3: Keep the relevant attributes

I only need the laboratory ID and the model ID.

$$\text{result} = \pi_{\text{labId}, \text{modelId}}(R_2)$$

## Step 4: Divide by the required categories

The division allows us to find the laboratories that are associated with all the required categories.

$$R_3 = \text{result} / \text{categories}$$

## Step 5: Project the laboratory IDs

Finally, I project the laboratory IDs.

$$\boxed{\pi_{\text{labId}}(R_3)}$$

Therefore, the final expression is:

$$\boxed{\pi_{\text{labId}}(\text{result} / \text{categories})}$$

where

$$\text{categories} = \pi_{\text{category}}(\sigma_{\text{category}=\text{"CPU Server"} \vee \text{category}=\text{"GPU Server"}}(\text{EquipmentModel}))$$

and

$$\text{result} = \pi_{\text{labId}, \text{modelId}}(\text{EquipmentUnit} \bowtie_{\text{modelId}} \text{EquipmentModel}).$$

## Question 5

Find the Faculty members who supervise every laboratory located in the building 'College of Computing'.

First, we select the rows from laboratory table that have college of computing as an attribute of building :

$$A = \text{building='college of computing'}(\text{laboratory})$$

Finally, we return the supervisorPid:

$$\boxed{\pi_{\text{supervisorPid}}(A)}$$

## Question 6

*Find the IDs of persons who participate in every research project funded by at least one budget that also funds the laboratory with labId = 2*

First, we select the laboratories that have labId= 2 from the table FundsLab:

$$A = \sigma_{\text{labId}=2}(\text{FundsLab})$$

Then, we join A with the table FundsPrj to get the project code :

$$B = A \bowtie \text{FundsPrj}$$

After that, we join B with the participants table :

$$C = B \bowtie \text{participants}$$

We return only the person ids:

$$\boxed{\pi_{\text{pid}}(C)}$$

## Question 7

*Find pairs of persons (p1, p2) such that p1 has made more reservations than p2.*

First, we count the reservation made by each person using the grouping aggregation in the Reservation table:

$$A = \gamma_{\text{madebyPid}}(n \leftarrow \text{COUNT}(\text{resId}))(\text{Reservation})$$

Then we make two copies of A by renaming the first copy C1 and the second copy C2 :

$$\begin{aligned} &\rho(C1(p1 \leftarrow \text{madebypid}, n \leftarrow n1), A) \\ &\rho(C2(p1 \leftarrow \text{madebypid}, n \leftarrow n2), A) \end{aligned}$$

Next, we join the two tables C1 and C2 based on a condition which is n1 bigger than n2 :

$$B = C1 \bowtie_{n1 > n2} C2$$

Finally, we keep only the person ids p1 and p2 :

$$\boxed{\pi_{p1,p2}(B)}$$

## Question 8

*Find the IDs of persons who have reserved equipment units located in at least two different laboratories.* First, we join Reserves with Reservation and Equipment unit :

$$A = Reserves \bowtie Reservation \bowtie Equipmentunit$$

Next, we select only the person ids and laboratory ids:

$$B = \pi_{pid,labId}(A)$$

Then, we count number of laboratory ids reserved by each person using grouping aggregation:

$$C = \gamma_{pid}(n \leftarrow COUNT(labId))(B)$$

Next, we select person ids who have n bigger than 2 from the table C :

$$D = \sigma_{n >= 2}(C)$$

Finally, we only select the person ids from the table D :

$$\boxed{\pi_{madebyPid}(D)}$$

## Question 9

*Find the IDs of reservations submitted in September 2026 that reserve at least one equipment unit currently located in the laboratory named 'AI Movement'.*

First, let us get a hold of the reservations that did take place along their respective equipment unit reserved:

$$C = (Reservation \bowtie_{resId} Reserves) \bowtie_{serialNo} EquipementUnit$$

Now, we also need to add the laboratories so that we can access the laboratory name.

$$A = C \bowtie_{labId} Laboratory$$

Let us filter out those equipment units that do not belong to a laboratory which name is "AI movement" and that have not been submitted in September 2026

$$\sigma_{name="AIMovement" \wedge (submissionTS < "2026-09-01 00:00:00" \wedge (submissionTS \geq "2026-09-01 00:00:00"))}(A)$$

Finally, we return the reservation IDs:

$$\pi_{resID} \left( \begin{array}{l} \sigma_{name = "AI Movement"} \\ \wedge (submissionTS < "2026 - 09 - 01 00 : 00 : 00") \\ \wedge submissionTS \geq "2026 - 09 - 01 00 : 00 : 00") \end{array} \right) \quad (A)$$

## Question 10

*Find the IDs of persons who act as PI of more than one research project*

Let us first get a hold of the people who play a "PI" role in "Participates" relation

$$R = \sigma_{role='Approved'}(Participates)$$

Used a grouping + aggregation method to get a hold of how many projects each person partook in, then filter in those who participated in one or more projects :

$$\sigma_{nbProjects>1}(\gamma_{pId; COUNT(code) \rightarrow nbProjects})(R)$$

Finally, we project the Person IDS

$$\pi_{pId}(\sigma_{nbProjects>1}(\gamma_{pId; COUNT(code) \rightarrow nbProjects})(R))$$

## Question 11

*Find the IDs of approvers who have approved reservations involving equipment units located in more than.*

First, we need a table that joins the reservations made with equipment units:

$$R = (Reserves \bowtie_{resId} Reservation) \bowtie_{serialNo} EquipementUnit$$

We will then use a projection to single out the labId attributes and get rid of duplicates:

$$A = \pi_{approvers, located}(R)$$

Finally, let's use grouping-aggregation to get a hold of the approvers with how many different labs equipment units they approved then we select those who select equipment units from different labs

$$\pi_{approversId}(\sigma_{nbDiffLabs>1}(\gamma_{approversId; COUNT(labId) \rightarrow nbDiffLabs}(A)))$$

## Question 12

*Find the IDs of persons who have no reservation whose planned reservation interval includes November 6, 2026.*

Let us first join between people and the reservations they made (we'll use an inner join to exclude the people who have made no reservations)

$$C = \text{People} \bowtie_{\text{People.pId}=\text{Reservation.madeById}} \text{Reservation}$$

We then get a hold of the IDs of people who did make a reservation with a planned reservation interval that includes November 6th, 2026 (here I made the assumption that people who do have a reservation means that their reservation was accepted, if the assumption is wrong, we can always just delete the selection operator)

$$B = \pi_{pId} \left( \begin{array}{c} \sigma_{\text{status} = \text{"accepted"}} \\ \wedge (\text{plannedStart} \leq \text{"2026-11-06"}) \\ \wedge (\text{submissionTS} \geq \text{"2026-11-06"}) \end{array} \right) \quad (C)$$

And since we want the people who HAVE NOT made these reservations:

$$\boxed{\pi_{pId}(\text{People}) - B}$$

## Question 14

*For each laboratory, return the person who has made the greatest number of distinct approved reservations involving equipment units located in that laboratory.*

First, we select only the approved reservations:

$$R = \sigma_{\text{status}=\text{'Approved'}}(\text{Reservation})$$

A reservation is associated with equipment units through the **Reserves** relation. We therefore join the approved reservations with **Reserves** and **EquipmentUnit**:

$$J = R \bowtie_{R.resId=\text{Reserves.resId}} \text{Reserves} \bowtie_{\text{Reserves.serialNo}=\text{EquipmentUnit.serialNo}} \text{EquipmentUnit}$$

We only need the laboratory ID, the person who made the reservation, and the reservation ID:

$$D = \pi_{labId, madeByPid, resId}(J)$$



The projection is important because the same reservation may reserve several equipment units in the same laboratory. Since relational algebra uses sets, the projection removes duplicate tuples. Thus, each  $(labId, madeByPid, resId)$  combination appears only once.

We then count the number of distinct reservations for each laboratory and person:

$$C = \gamma_{labId, madeByPid; COUNT(resId) \rightarrow n}(D)$$

Next, we find the maximum number of reservations for each laboratory:

$$M = \gamma_{labId; MAX(n) \rightarrow maxN}(C)$$

We combine  $C$  and  $M$  by matching the laboratory and the maximum count:

$$W = C \bowtie_{C.labId=M.labId \wedge C.n=M.maxN} M$$

Finally, we return the laboratory ID and the ID of the person who has the greatest number of approved reservations:

$$\boxed{\pi_{labId, madeByPid}(W)}$$

## Question 15

*List the IDs of persons who made at least three equipment reservations during the year 2025.*

First, we select the reservations whose planned start time is during the year 2025:

$$R = \sigma_{plannedStart \geq '2025-01-01' \wedge plannedStart < '2026-01-01'}(Reservation)$$

We then count the number of reservations made by each person:

$$C = \gamma_{madeByPid; COUNT(resId) \rightarrow n}(R)$$

Next, we keep only the persons who made at least three reservations:

$$C' = \sigma_{n \geq 3}(C)$$

Finally, we return only their IDs:

$$\boxed{\pi_{madeByPid}(C')}$$

## Key Challenges

The main challenges in these queries were grouping and aggregation, Comparing the total values and dealing with separate reservations.

For Question 7, we had a problem regarding the output that is supposed to be two fields (p1,p2) which are not stated in two different tables. After pondering it, we realized that we could create two copies of the same table and then join them together.

For Question 8 and 11, we were confused about the COUNT aggregation function that will

count the same labid twice if the person reserved two equipment units located in the same lab. However, we cleared the confusion by using projection on labid and pid so that duplicate rows could be eliminated.